

Experiment:2 Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

Aim: To simulate and analyze the performance of a microwave transmission line and two-port network using scattering parameters (S-parameters), particularly S11 and S21.

Apparatus:

- Computer/Laptop
- MATLAB software

Procedure:

1. Open MATLAB and create a new script.
2. Define the frequency range from 1 GHz to 10 GHz.
3. Define the characteristic impedance and transmission-line length.
4. Calculate the propagation constant.
5. Calculate the scattering parameters S11 and S21.
6. Convert the S-parameters into dB using $(20 \log_{10} |S|)$.
7. Plot S21 versus frequency to observe transmission performance.
8. Plot S11 versus frequency to observe reflection performance.
9. Analyze the obtained plots.

Code:

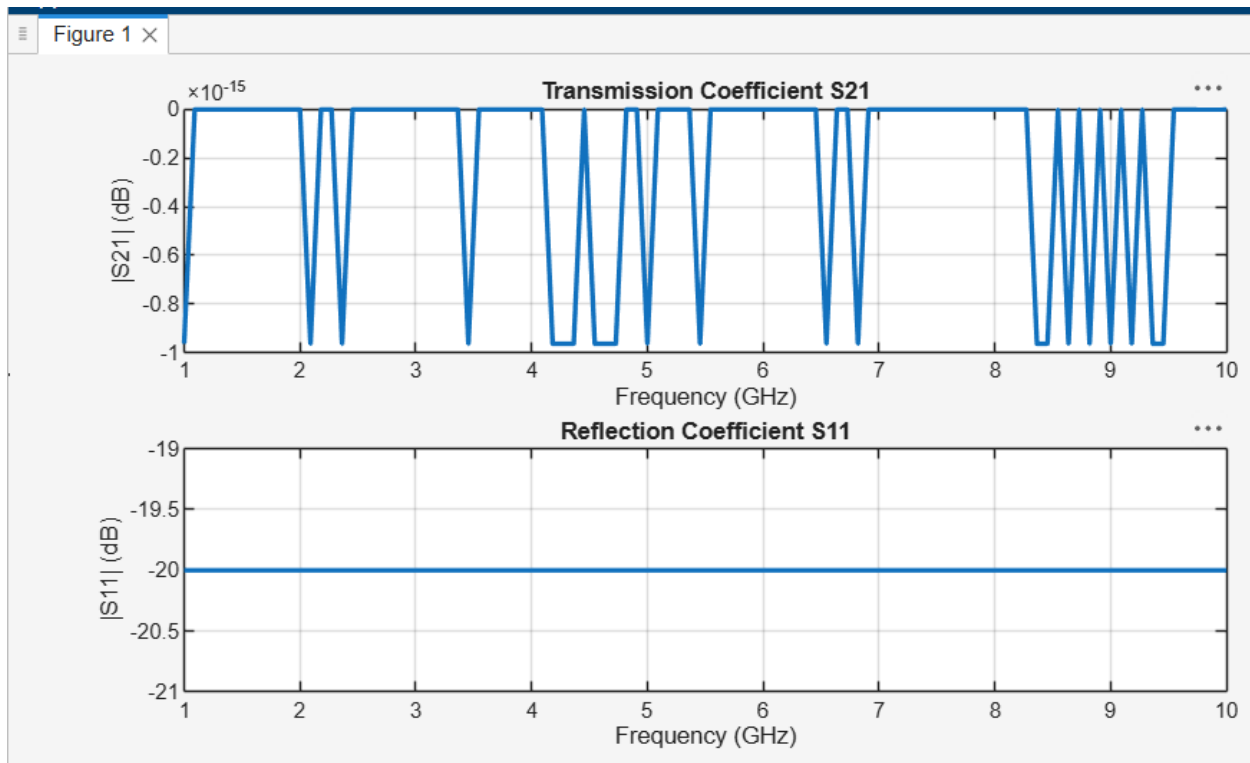
```
clc;
clear;
close all;
f = linspace(1e9,10e9,100);
c = 3e8;
Z0 = 50;
L = 10e-9;
beta = 2*pi*f/c;
S21 = exp(-1j*beta*L);
S11 = 0.1*ones(size(f));
S21_dB = 20*log10(abs(S21));
S11_dB = 20*log10(abs(S11));
figure;
```

```

subplot(2,1,1);
plot(f/1e9,S21_dB,'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('|S21| (dB)');
title('Transmission Coefficient S21');
subplot(2,1,2);
plot(f/1e9,S11_dB,'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('|S11| (dB)');
title('Reflection Coefficient S11');

```

Output:



Result:

The S11 and S21 parameters were successfully simulated, showing the reflection and transmission characteristics of the microwave network.